

Session 12: Structures Topics: Declaring, initializing structures. Nested structures. Example: Store student details (name, marks, grade).

Tasks

- 1) Declare a structure named Playlist to store details of a song: title (string), artist (string), and duration in seconds (integer). Initialize one Playlist variable with your favorite song's details and print each field.**

Code-

```
#include <stdio.h>
#include <string.h>

struct Playlist {
    char title[50];
    char artist[50];
    int duration;
};

int main() {

    struct Playlist mySong = {"Blinding Lights", "The Weeknd", 200};

    printf("Title: %s\n", mySong.title);
    printf("Artist: %s\n", mySong.artist);
    printf("Duration: %d seconds\n", mySong.duration);

    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc struct.c -o struct.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\struct.exe
Title: Blinding Lights
Artist: The Weeknd
Duration: 200 seconds
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

2) Create a structure called Food Item to store Zomato-style menu data: item Name (string), price (float), and rating (float). Initialize an array of 3 Food Item variables with real menu items and display their details using a loop.

Code-

```
#include <stdio.h>
```

```
struct FoodItem {  
    char itemName[50];  
    float price;  
    float rating;  
};
```

```
int main() {  
    struct FoodItem menu[3] = {  
        {"Butter Chicken", 320.00, 4.5},  
        {"Paneer Tikka", 250.00, 4.3},  
        {"Veg Biryani", 180.00, 4.1}  
    };  

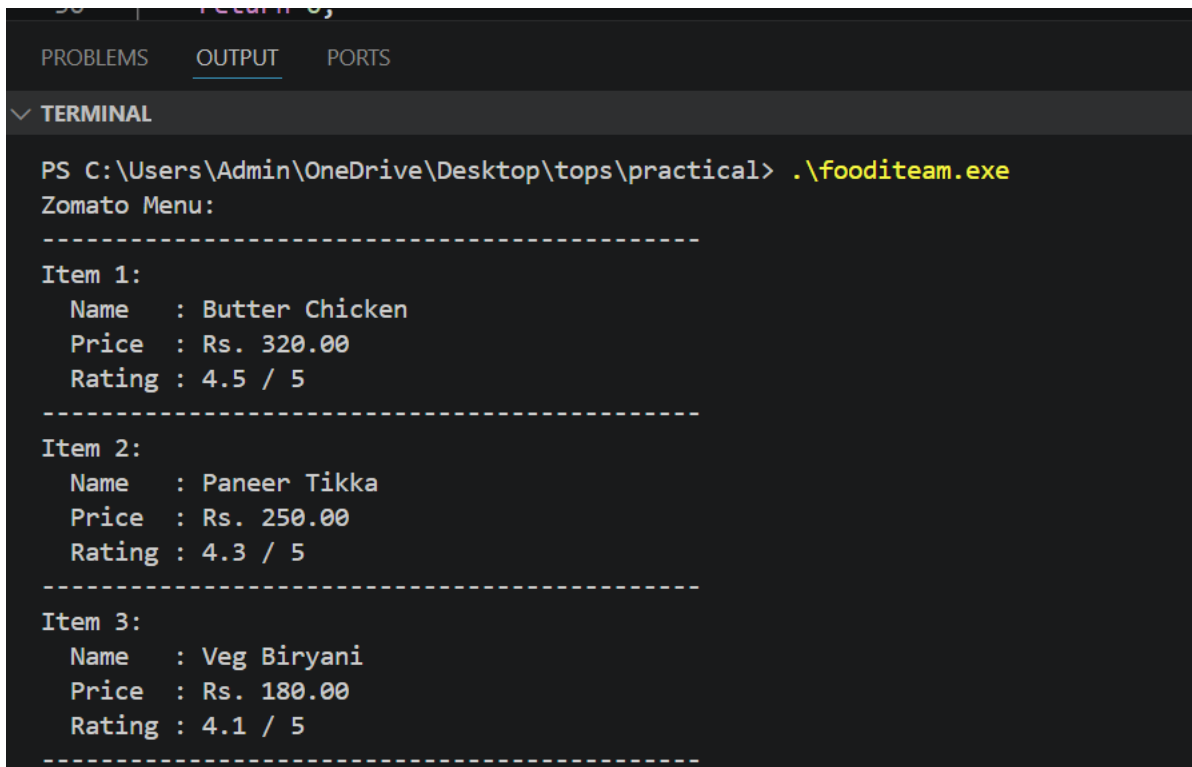
```

```
    printf("Zomato Menu:\n");  
    printf("-----\n");
```

```
    for (int i = 0; i < 3; i++) {  
        printf("Item %d:\n", i + 1);
```

```
        printf(" Name   : %s\n", menu[i].itemName);  
        printf(" Price  : Rs. %.2f\n", menu[i].price);  
        printf(" Rating : %.1f / 5\n", menu[i].rating);  
        printf("-----\n");  
    }  
  
    return 0;  
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS  
✓ TERMINAL  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\fooditeam.exe  
Zomato Menu:  
-----  
Item 1:  
  Name   : Butter Chicken  
  Price  : Rs. 320.00  
  Rating : 4.5 / 5  
-----  
Item 2:  
  Name   : Paneer Tikka  
  Price  : Rs. 250.00  
  Rating : 4.3 / 5  
-----  
Item 3:  
  Name   : Veg Biryani  
  Price  : Rs. 180.00  
  Rating : 4.1 / 5  
-----
```

3) Define a nested structure called Movie Show for a Book My Show-style app: Movie (string), Screen (integer), and a nested structure Time with hours and minutes (integers). Create and initialize a Movie Show variable for any movie and print its details in the format 'Movie: X, Screen: Y, Time: HH:MM'.

Code-

```
#include <stdio.h>

struct Time {
    int hours;
    int minutes;
};

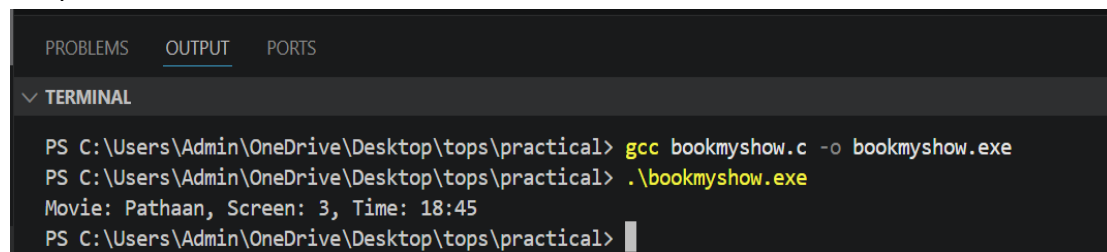
struct MovieShow {
    char movie[50];
    int screen;
    struct Time showTime;
};

int main() {
    struct MovieShow show1 = {"Pathaan", 3, {18, 45}};

    printf("Movie: %s, Screen: %d, Time: %02d:%02d\n",
        show1.movie, show1.screen, show1.showTime.hours, show1.showTime.minutes);

    return 0;
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
▼ TERMINAL
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc bookmyshow.c -o bookmyshow.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\bookmyshow.exe
Movie: Pathaan, Screen: 3, Time: 18:45
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> |
```

4) Build a structure called Insta Profile with fields: username (string), followers (integer), and a nested structure Bio with fields: description (string) and age (integer). Initialize an Insta Profile variable with your own details and display all fields

Code-

```
#include <stdio.h>
```

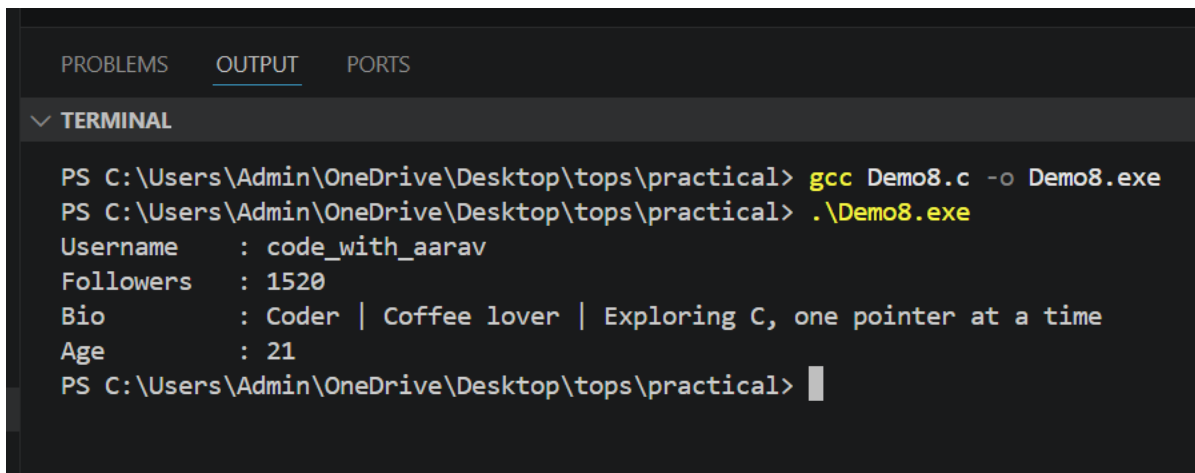
```
struct Bio {  
    char description[100];  
    int age;  
};
```

```
struct InstaProfile {  
    char username[50];  
    int followers;  
    struct Bio bio;  
};
```

```
int main() {  
    struct InstaProfile myProfile = {  
        "code_with_aarav",  
        1520,  
        {"Coder | Coffee lover | Exploring C, one pointer at a time", 21}  
    };
```

```
printf("Username   : %s\n", myProfile.username);  
printf("Followers  : %d\n", myProfile.followers);  
printf("Bio       : %s\n", myProfile.bio.description);  
printf("Age       : %d\n", myProfile.bio.age);  
  
return 0;  
}
```

Output:



```
PROBLEMS  OUTPUT  PORTS
✓ TERMINAL

PS C:\Users\Admin\OneDrive\Desktop\tops\practical> gcc Demo8.c -o Demo8.exe
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\Demo8.exe
Username    : code_with_aarav
Followers   : 1520
Bio         : Coder | Coffee lover | Exploring C, one pointer at a time
Age         : 21
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> 
```